

Rainband-Occurrence Probability in Northern Hemisphere Tropical Cyclones by Synthetic Aperture Radar Imagery

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Introduction

Figure S1 shows the rainbands in SAR TC images and the collocated SFMR rain rate data. Figure S2 shows the different types of rainbands in SAR TC images. The results of rainband annotation are shown in Figure S3. Figure S4 is a flow chart of the rainband-occurrence probability estimation. Figure S5 shows the diagram of the coordinate systems used in this study. Figures S6a to S6f show the Northern Hemisphere early morning TCs/evening TCs rainband-occurrence probability difference under different TC intensities and ocean basins. At the same time, the Northern Hemisphere strong TCs/weak TCs rainband-occurrence probability difference is shown in Figure S6g. Figures S7 and S8 show the variations of rainband-occurrence probability with radial distance in Figure 3 and its downshear region, respectively. Figure S9 shows the variation of rainband-occurrence probability with azimuth in Figure 3. Figures S10 and S11 show the variations of rainband-occurrence probability with radial distance in Figure 4 and its downshear region, respectively. Figure S12 shows the variation of rainband-occurrence probability with azimuth in Figure 4. Figure S13 gives the histograms of the distances and MWR-normalized distances between the BT-interpolated and manually-recognized TC center positions in Text S1. Figures S14 to S19 provide the results of the automatically extracted rainbands by the U-Net deep learning model in Text S2.

Table S1 shows the numbers of data in different classes. Table S2 gives the medians of the rainband-occurrence probability in the early morning and evening for TCs in the three ocean basins. Table S3 shows the result of the Mann-Whitney U test among the rainband-occurrence probabilities in the three ocean basins. Table S4 provides the statistics of the distances and MWR-normalized distances between the BT-interpolated and manually-recognized TC center positions in Text S1.

Text S1 compares the BT-interpolated and manually-recognized TC center positions. Text S2 extracts the rainbands automatically with the U-Net deep learning model.

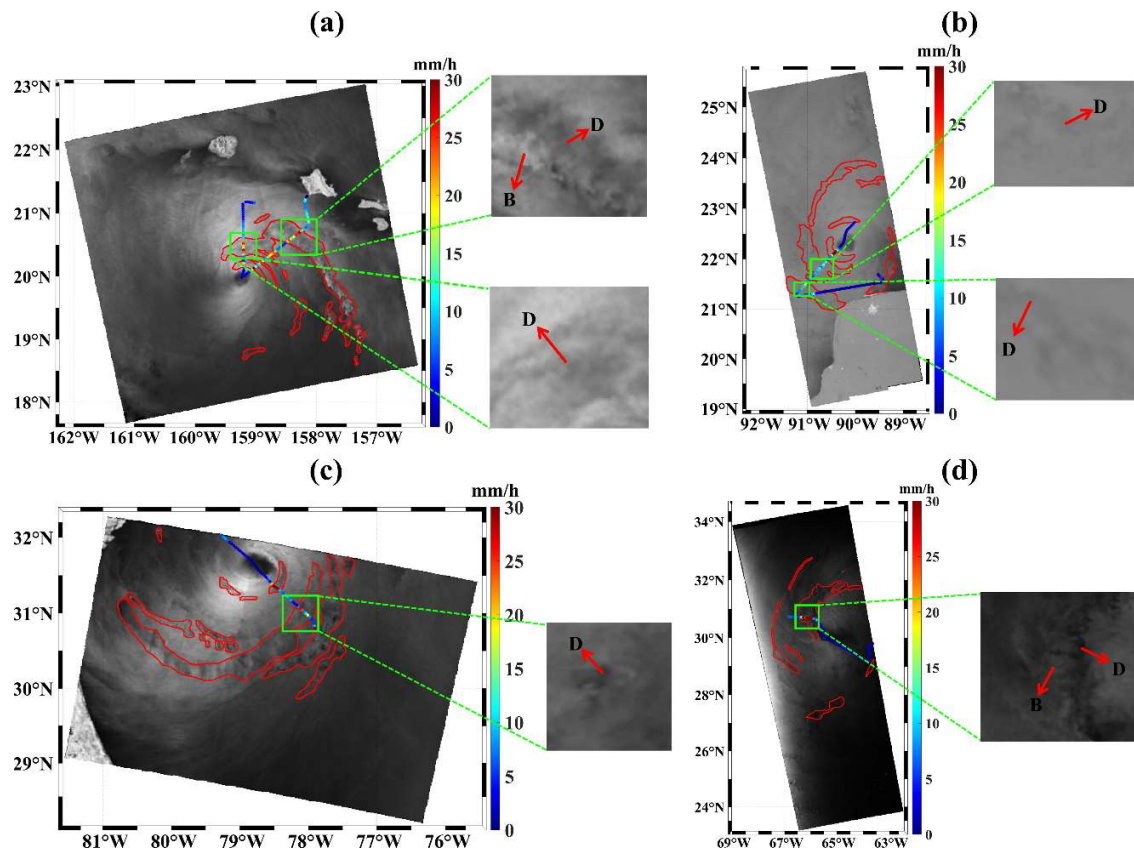


Figure S1. The rainbands (enclosed by red lines) in SAR TC images and the collocated SFMR rain rate data. Letters "B" and "D" stand for "Bright" and "Dark", respectively. (a) TC Ana at 04:44:31 UTC on 19 October 2014. (b) TC Delta at 00:07:02 UTC on 8 October 2020. (c) TC Arthur at 11:13:56 UTC on 3 July 2014. (d) TC Karl at 22:21:32 UTC on 23 September 2016.

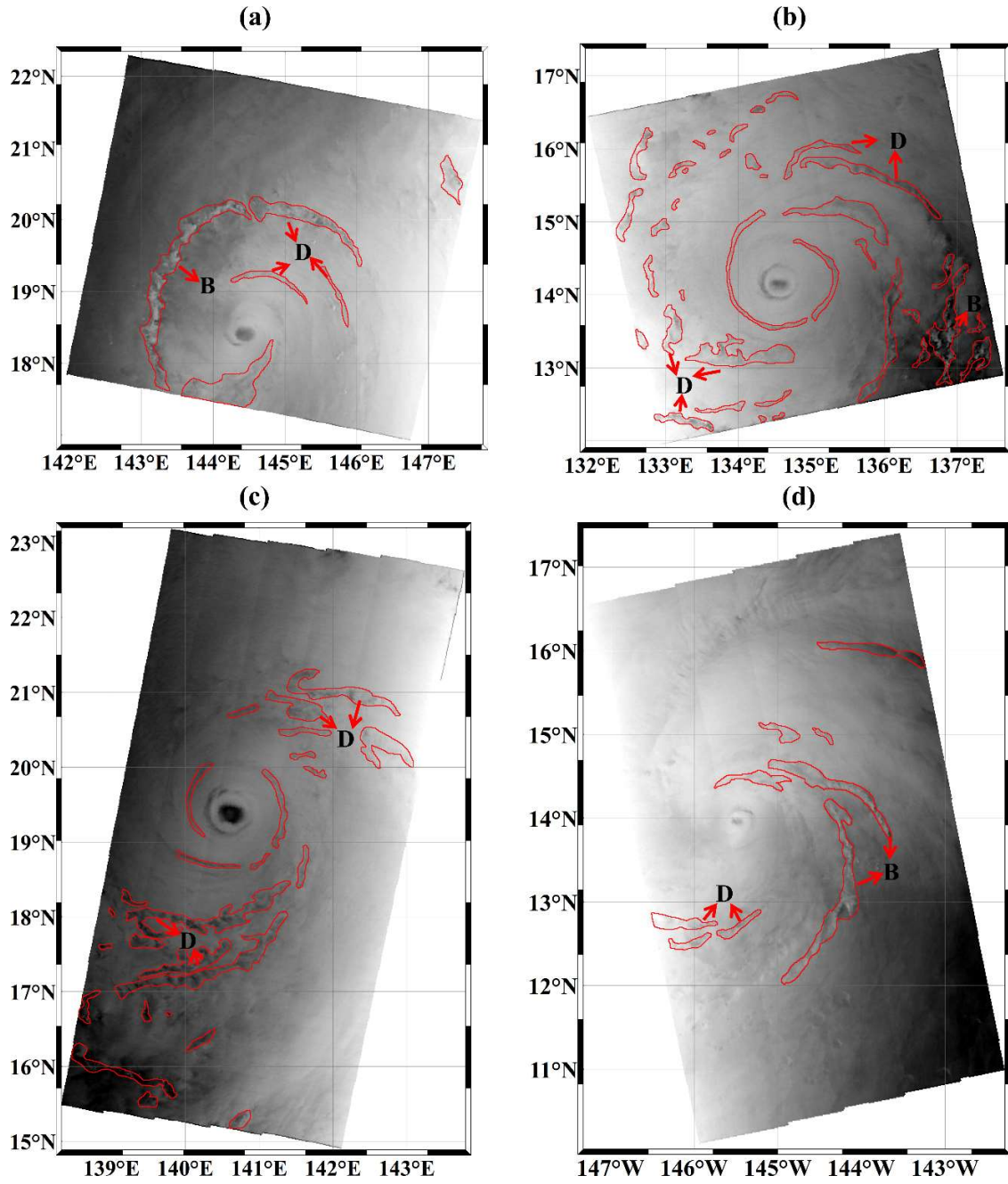


Figure S2. The rainbands observed in SAR TC images. (a) TC Fengshen at 20:12:29 UTC on 14 November 2019. (b) TC Mangkhut at 09:10:24 UTC on 12 September 2018. (c) TC Hagibis at 20:30:40 UTC on 8 October 2019. (d) TC Erick at 03:47:21 UTC on 31 July 2019. The rainbands are annotated by the red-line enclosed areas.

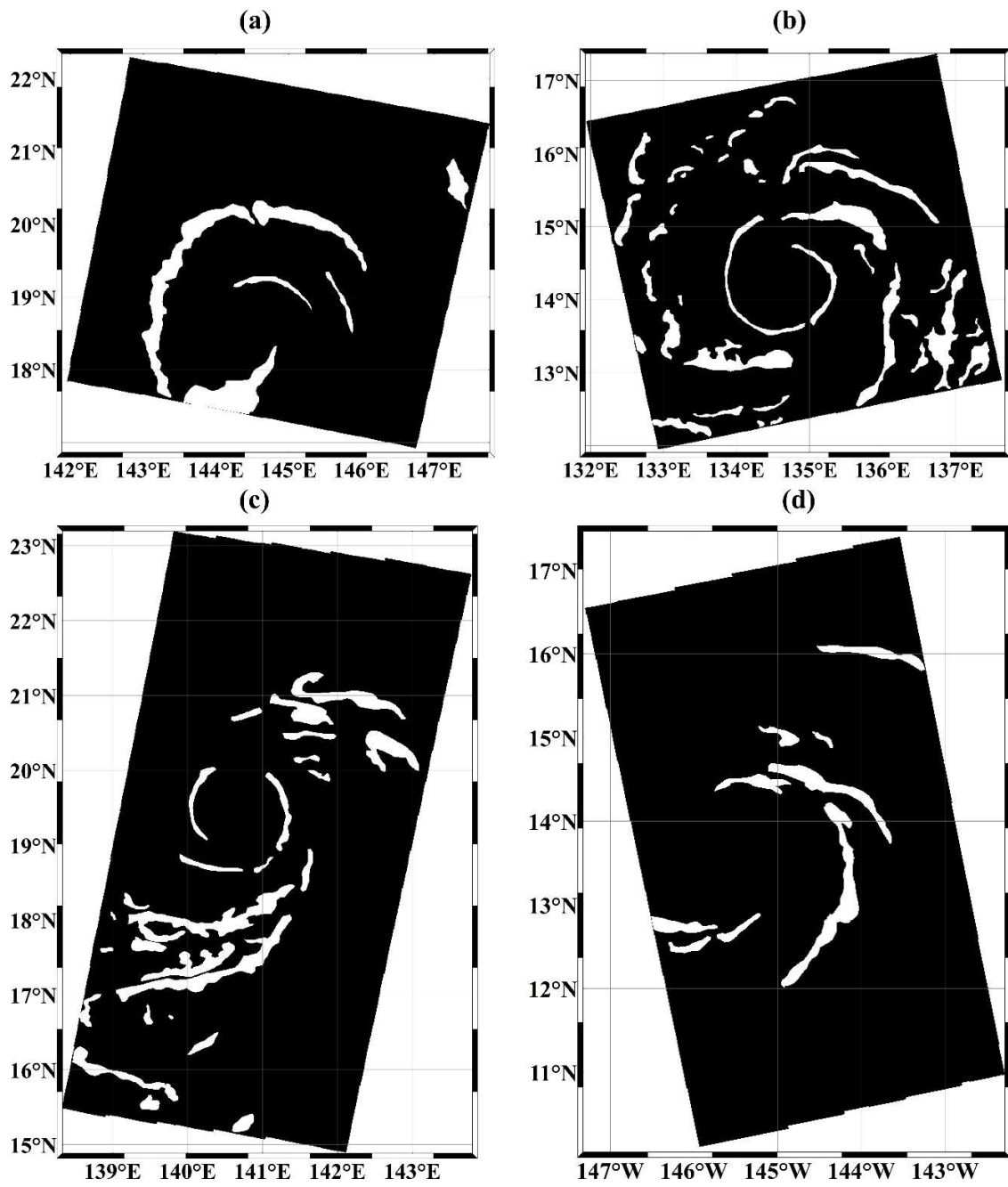


Figure S3. The results of rainband annotation of SAR TC images in Figure S2.

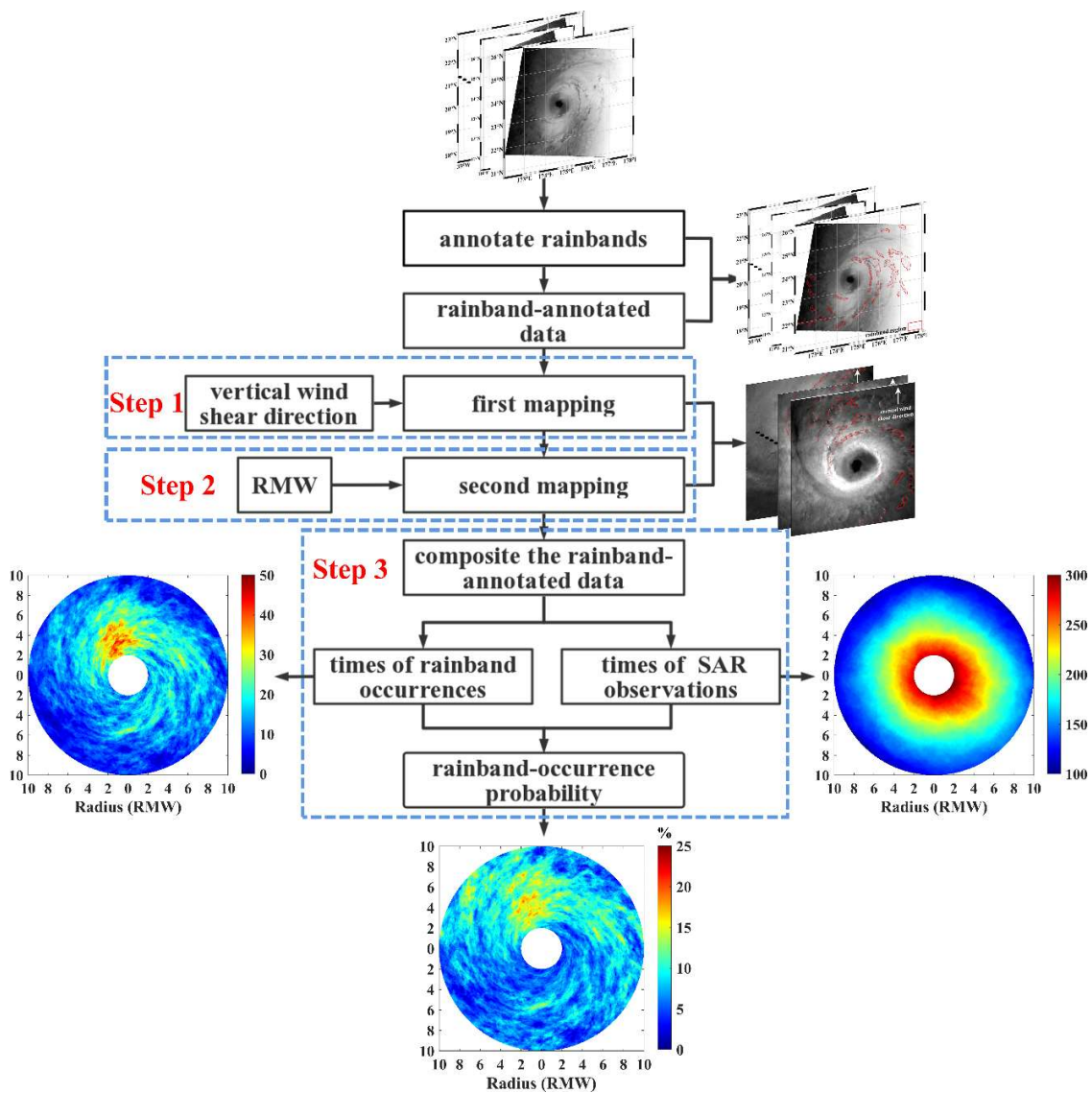


Figure S4. The flow chart of the rainband-occurrence probability estimation.

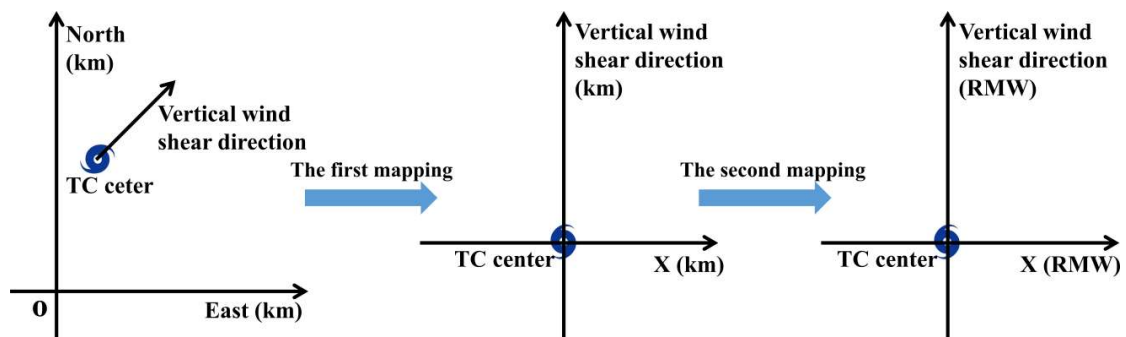


Figure S5. Coordinate system diagram.

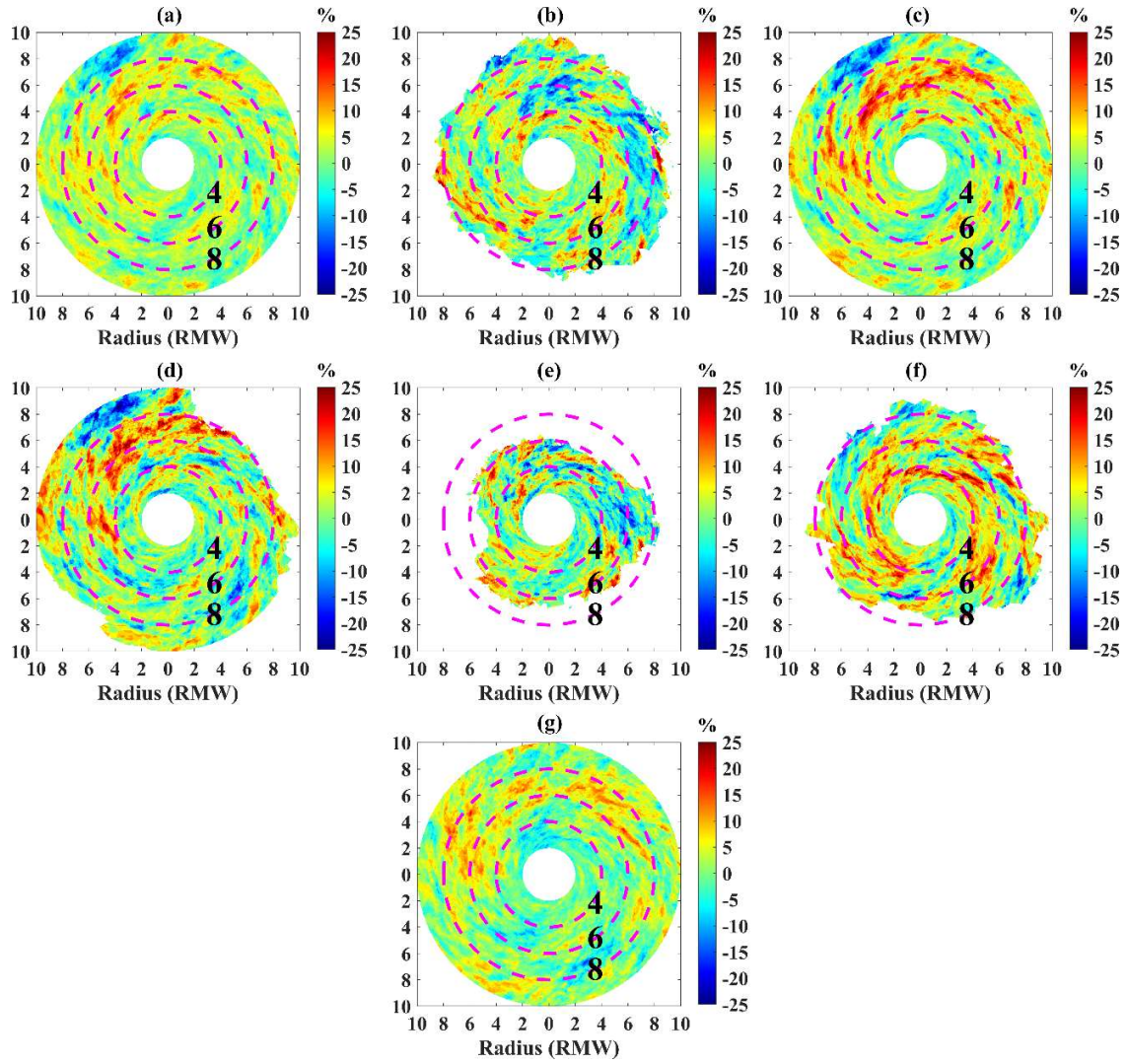


Figure S6. Rainband-occurrence probability difference. (a)-(f) are the early morning TCs/evening TCs rainband-occurrence probability difference in (a) Northern Hemisphere TCs, (b) Northern Hemisphere weak TCs, (c) Northern Hemisphere strong TCs, (d) NWP TCs, (e) NA TCs, and (f) NEP TCs. (g) is the strong TCs/weak TCs rainband-occurrence probability difference in Northern Hemisphere TCs.

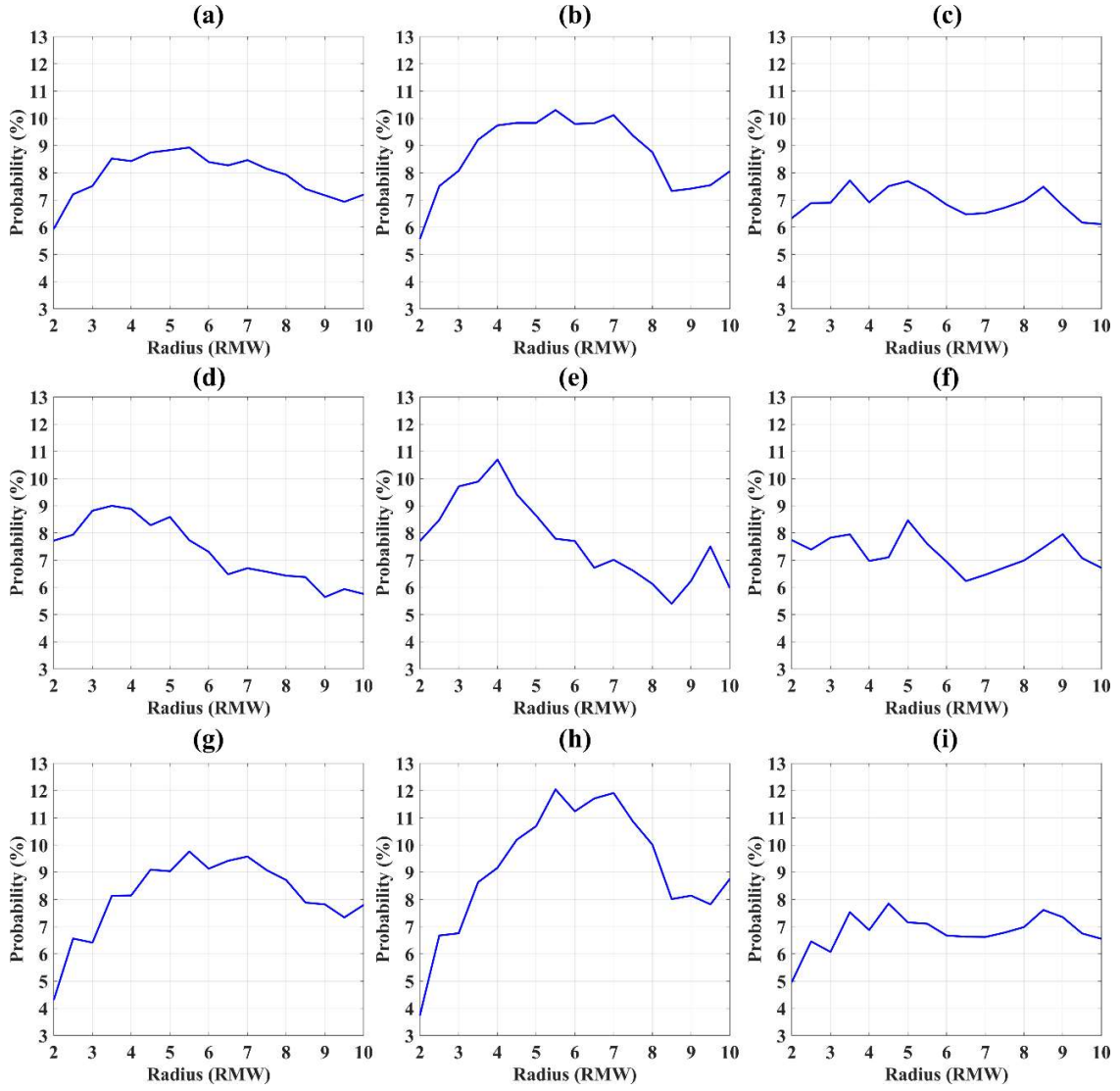


Figure S7. The variation of the Northern Hemisphere rainband-occurrence probability with radial distance (corresponding to Figure 3).

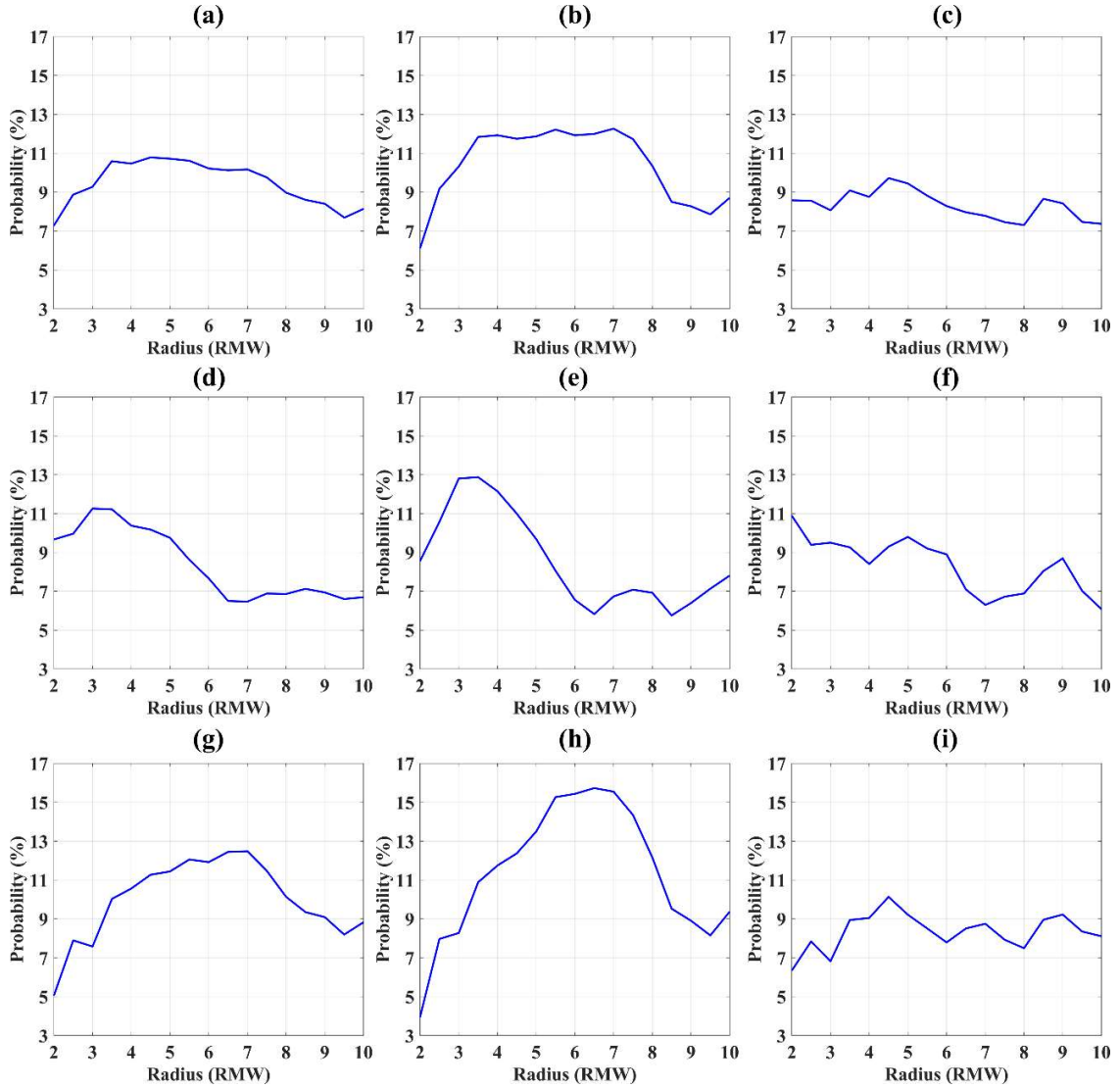


Figure S8. The variation of the Northern Hemisphere rainband-occurrence probability with radial distance in the downshear region of TCs (corresponding to Figure 3).

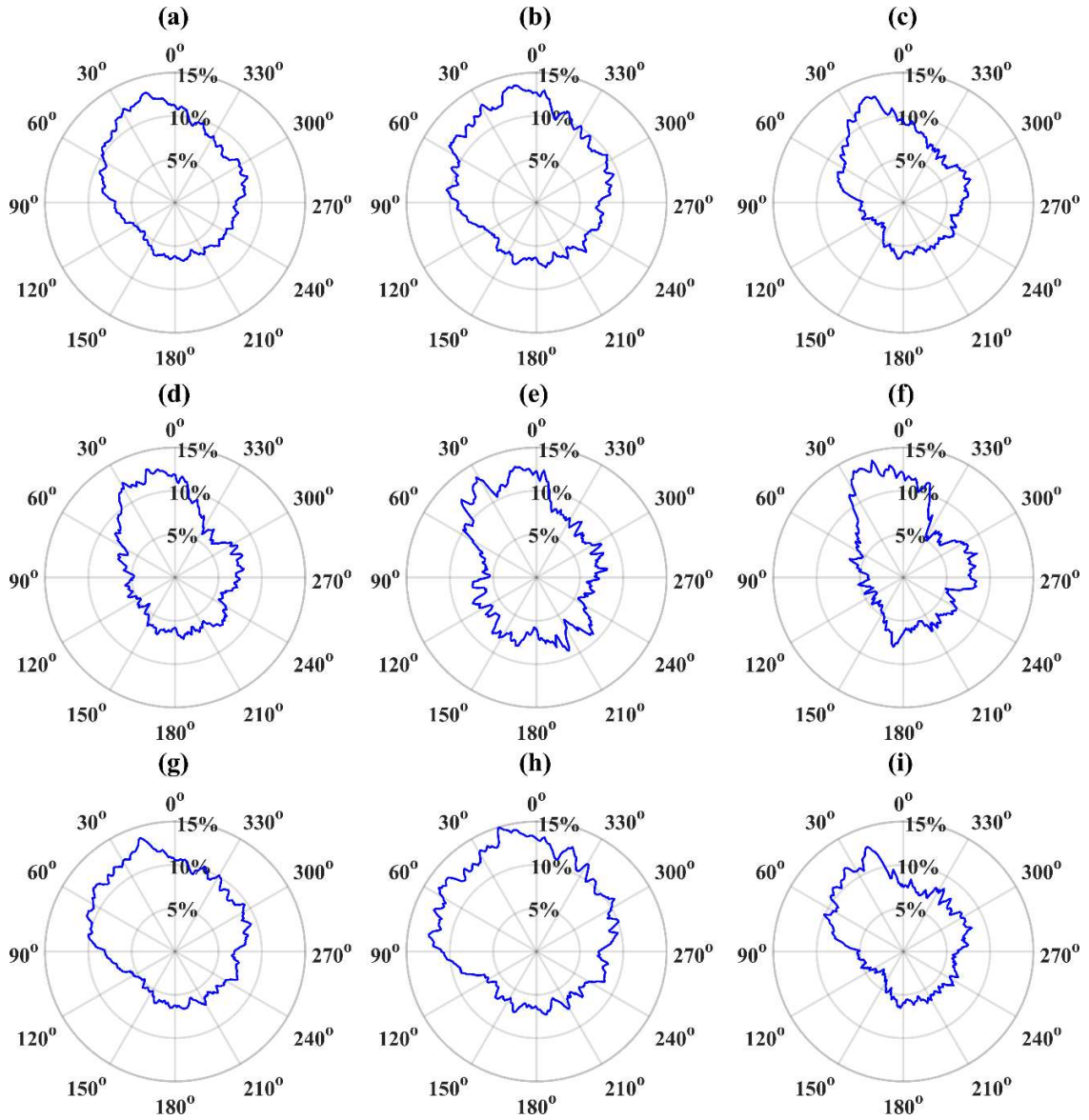


Figure S9. The variation of the Northern Hemisphere rainband-occurrence probability with azimuth (corresponding to Figure 3).

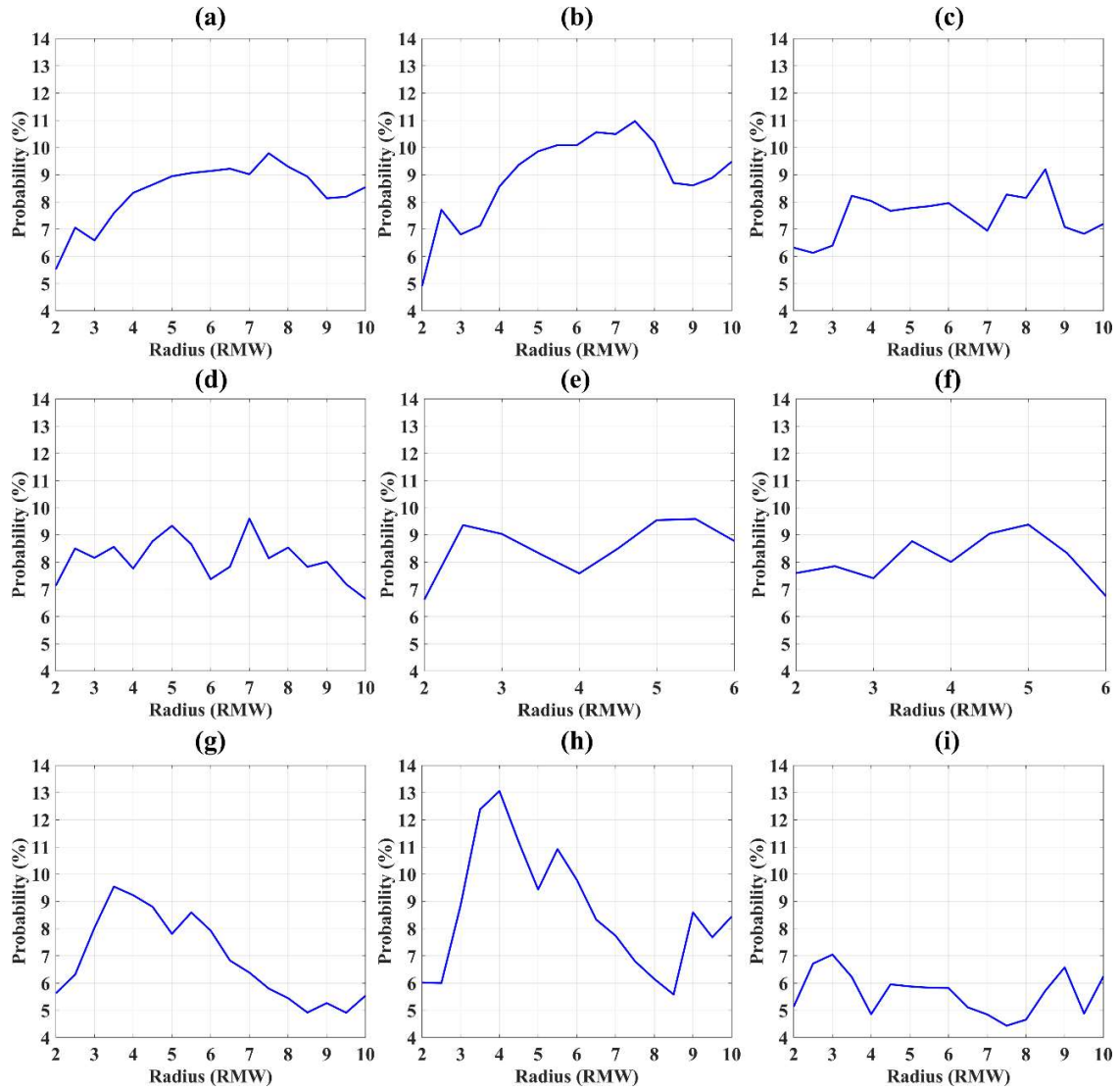


Figure S10. The variation of the Northern Hemisphere rainband-occurrence probability with radial distance (corresponding to Figure 4).

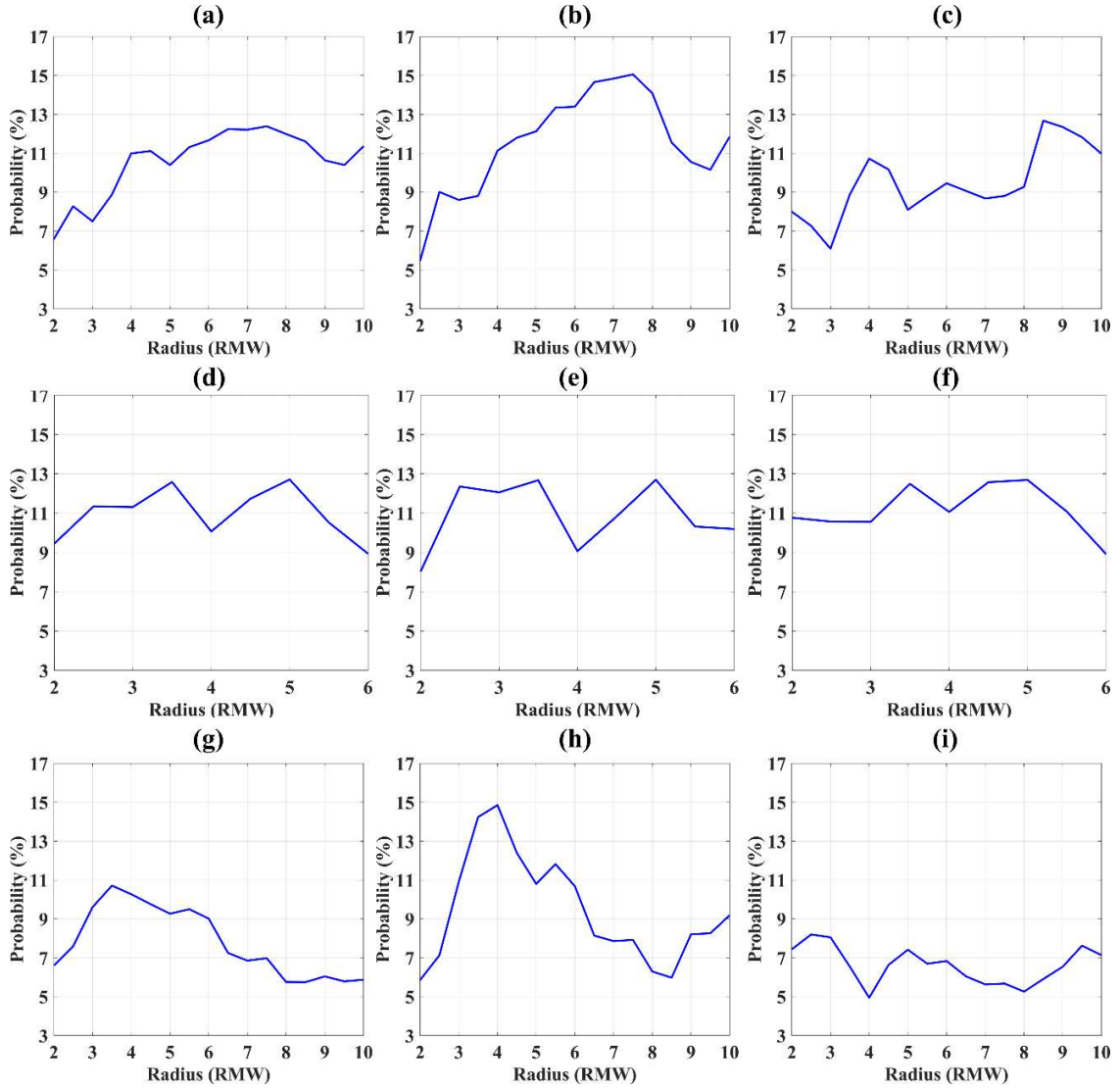


Figure S11. The variation of the Northern Hemisphere rainband-occurrence probability with radial distance in the downshear region of TCs (corresponding to Figure 4).

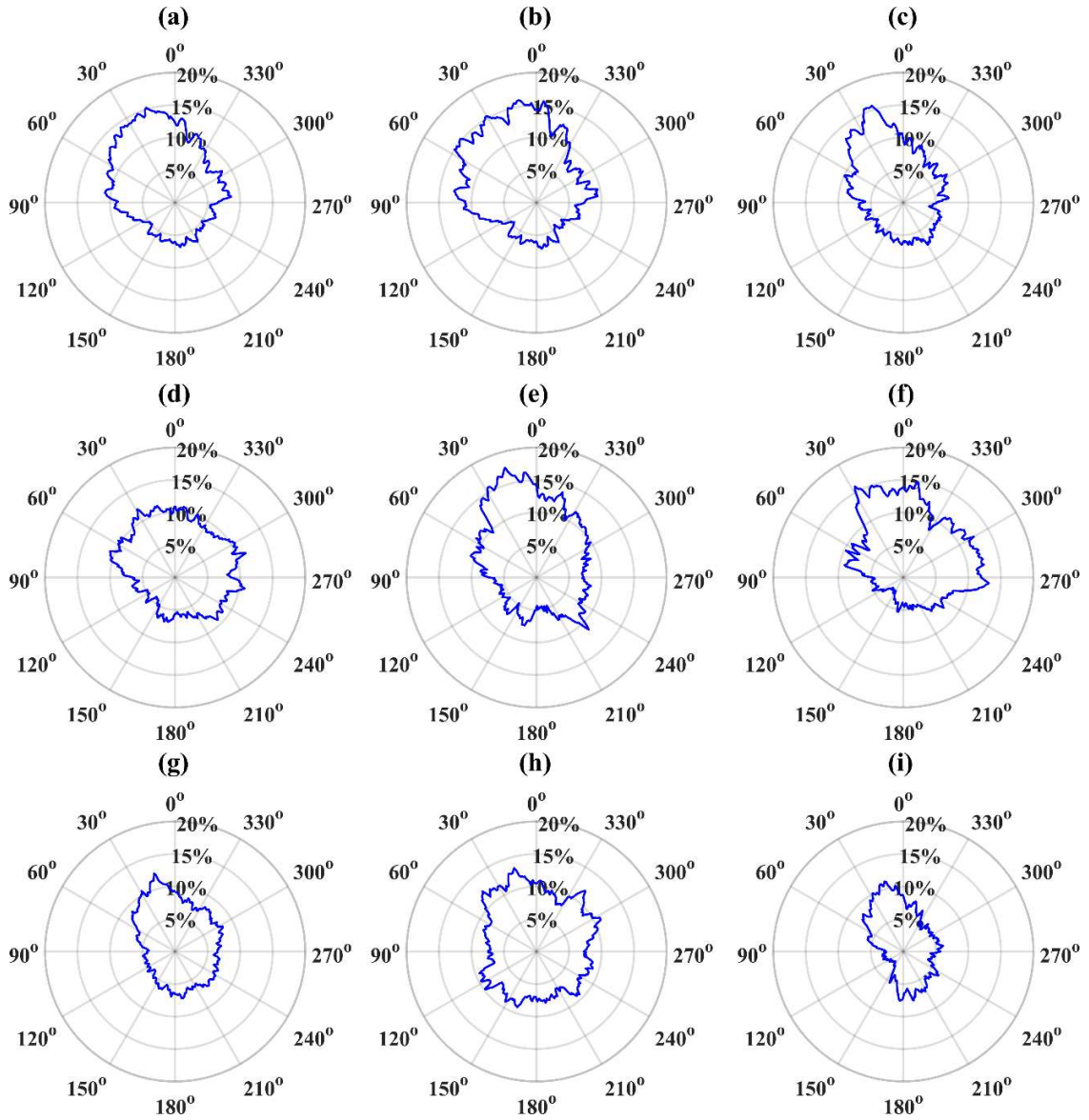


Figure S12. The variation of the Northern Hemisphere rainband-occurrence probability with azimuth (corresponding to Figure 4).

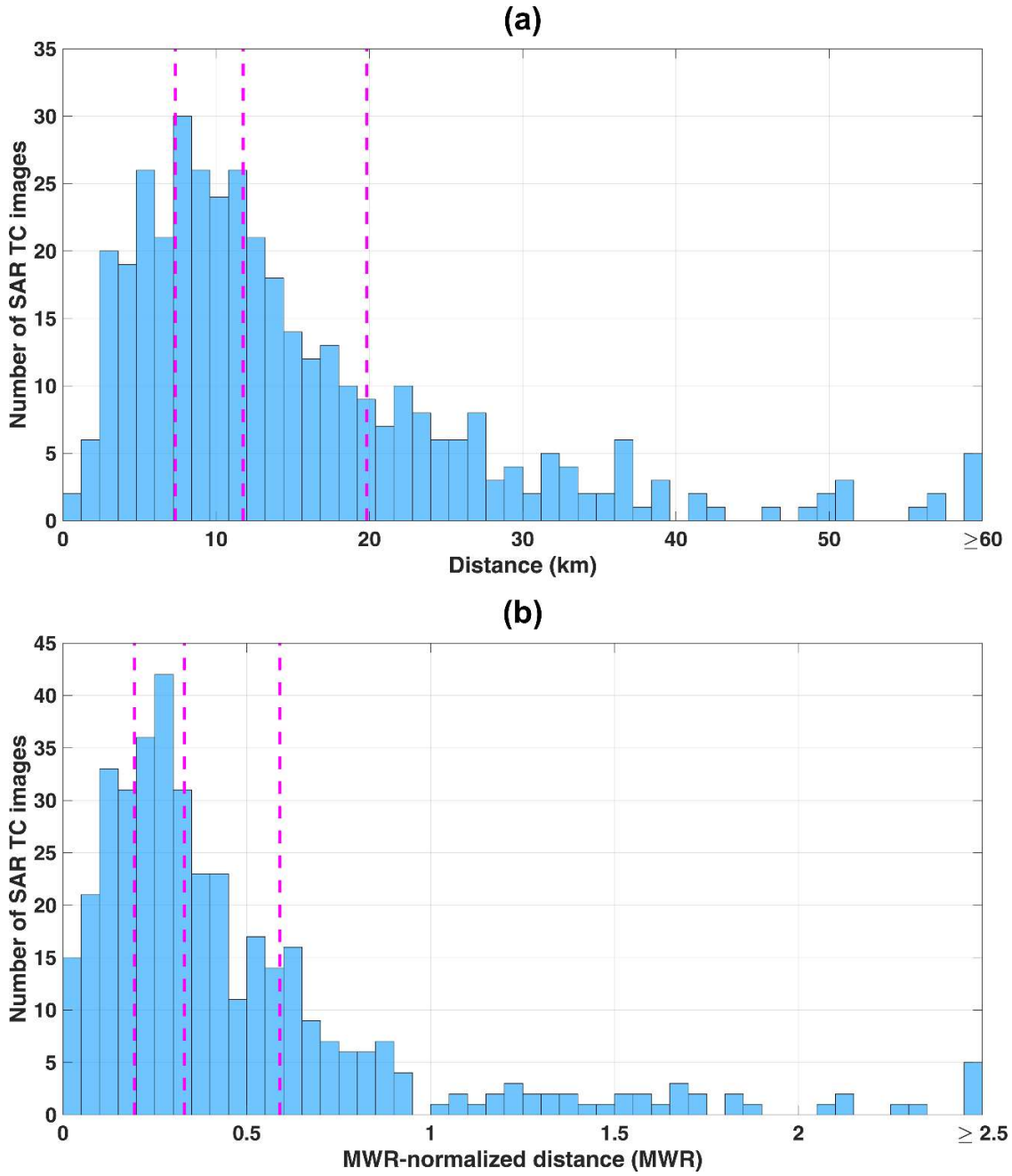


Figure S13. The histograms of (a) the distances and (b) MWR-normalized distances between the BT-interpolated and manually-recognized TC center positions, where magenta lines denote the positions of 25% quartile, 50% quartile (median), and 75% quartile.

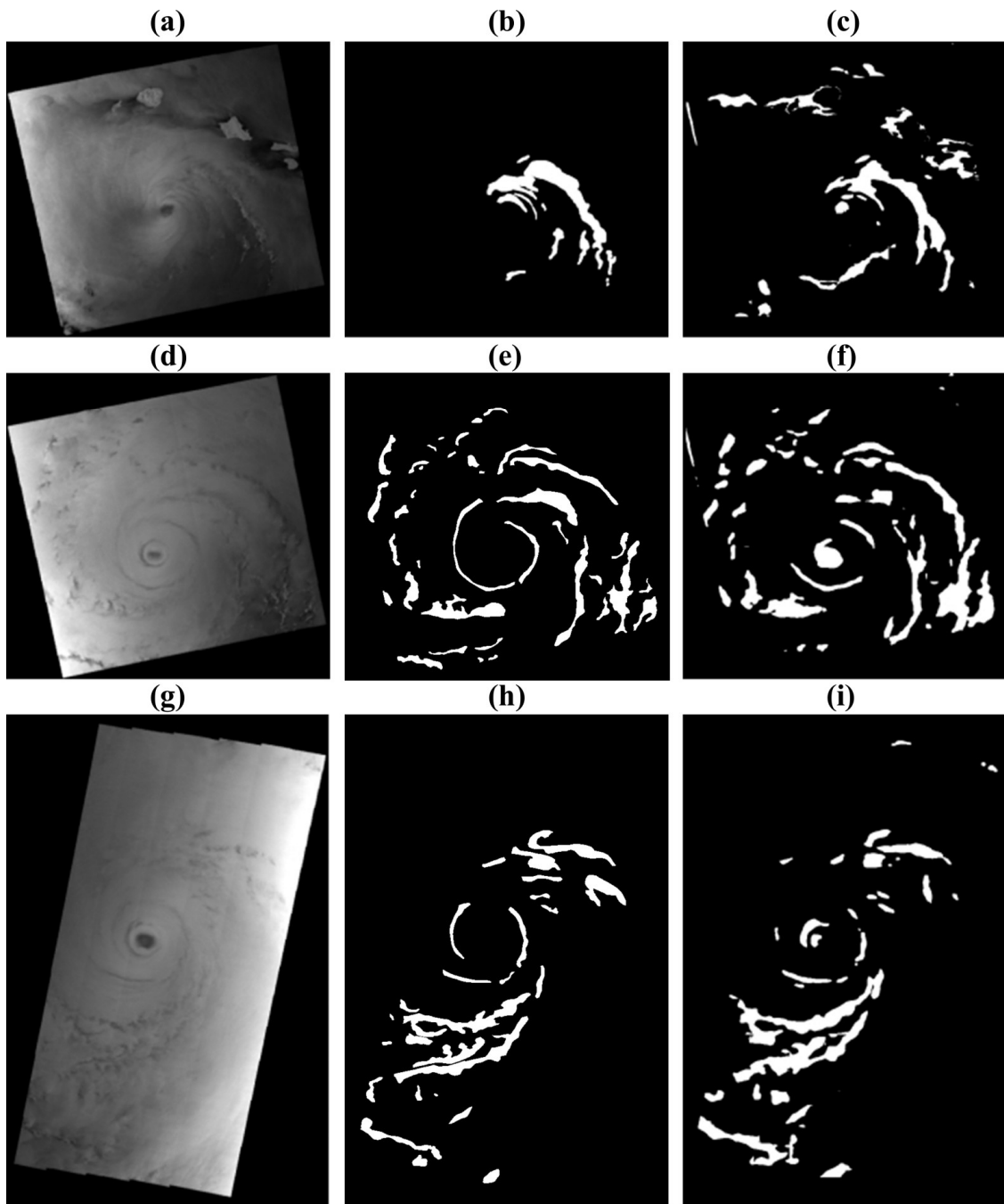


Figure S14. Rainbands' manual annotation and automatic extraction results. The first column is the SAR TC images ((a), (d), and (g)). The second column is the manual annotation result ((a), (d), and (g)). The third column is the automatic extraction result by the U-Net model ((c), (f), and (i)).

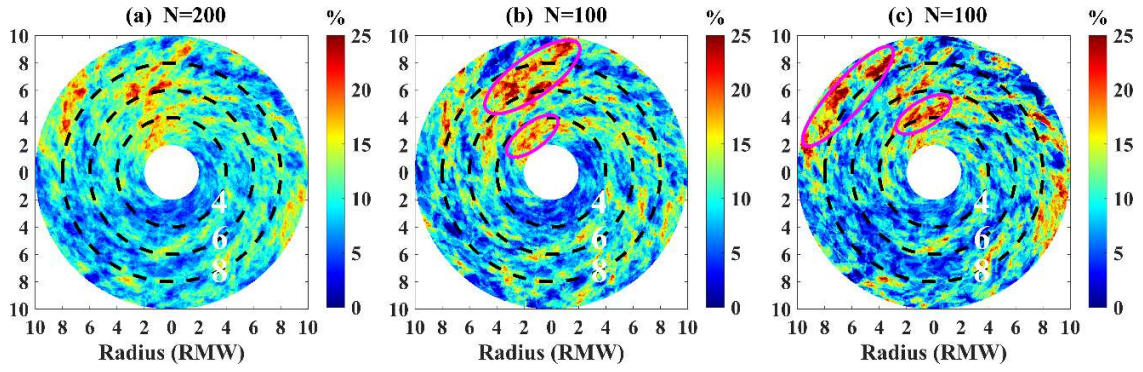


Figure S15. The rainband-occurrence probability of Northern Hemisphere TCs based on rainbands automatically extracted by the U-Net model. (a) Northern Hemisphere TCs, (b) Northern Hemisphere TCs in the early morning, and (c) Northern Hemisphere TCs in the evening. The peak region is marked with ellipses.

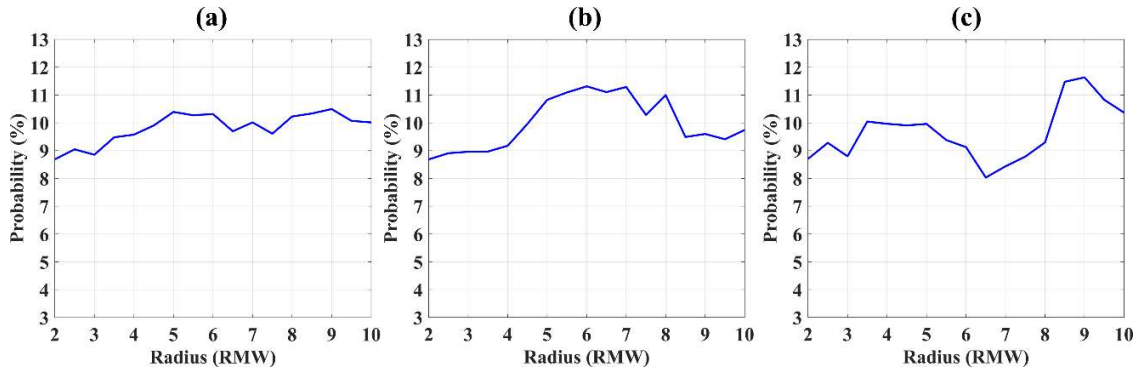


Figure S16. The radial variation of the Northern Hemisphere rainband-occurrence probability based on rainbands automatically extracted by the U-Net model. (a) Northern Hemisphere TCs, (b) Northern Hemisphere TCs in the early morning, and (c) Northern Hemisphere TCs in the evening.

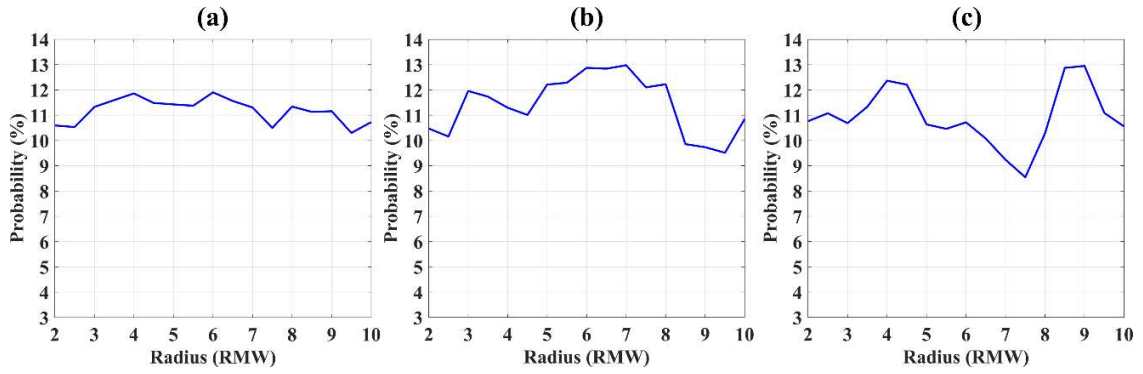


Figure S17. The radial variation of the Northern Hemisphere rainband-occurrence probability in the downshear region of TCs based on rainbands automatically extracted by the U-Net model. (a) Northern Hemisphere TCs, (b) Northern Hemisphere TCs in the early morning, and (c) Northern Hemisphere TCs in the evening.

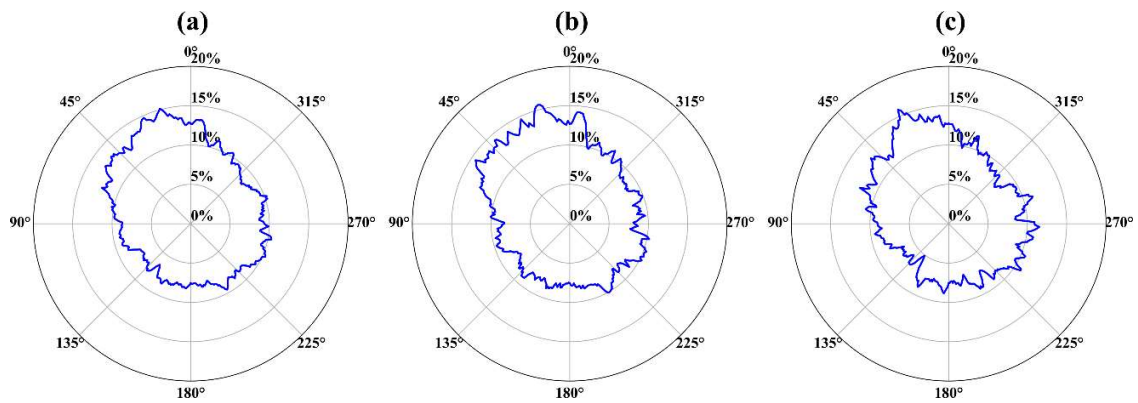


Figure S18. The azimuth variation of the Northern Hemisphere rainband-occurrence probability based on rainbands automatically extracted by the U-Net model. (a) Northern Hemisphere TCs, (b) Northern Hemisphere TCs in the early morning, and (c) Northern Hemisphere TCs in the evening.

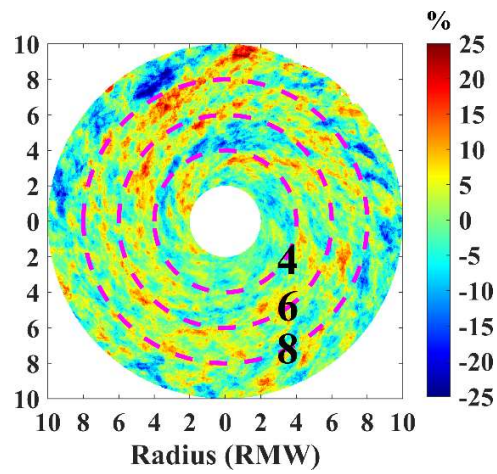


Figure S19. The early morning TCs/evening TCs rainband-occurrence probability difference in Northern Hemisphere TCs, based on rainbands automatically extracted by the U-Net model.

(a)							
	NWP	NA	NEP	NIO	SIO	SP	Total number
Local early morning	76	57	57	4	44	4	242
Local evening	60	72	53	4	29	4	222
Total number	136	129	110	8	73	8	464

(b)				
	NWP	NA	NEP	Total number
Tropical Storm and Category 1	51	73	61	185
Categories 2 to 5	85	56	49	190
Total number	136	129	110	375

Table S1. The numbers of data in different classes. (a) Different ocean basins and local times, (b) different ocean basins and TC intensities. NIO, SIO, and SP stand for the North Indian Ocean, the South Indian Ocean, and the South Pacific, respectively.

(a)			
	NWP	NA	NEP
Local early morning	8.33%	8.33%	8.33%
Local evening	6.90%	7.41%	4.35%

(b)			
	NWP	NA	NEP
Local early morning	11.76%	9.76%	9.09%
Local evening	9.09%	10.81%	6.06%

Table S2. The medians of the rainband-occurrence probability in the whole (a) and downshear (b) regions of TCs in different ocean basins.

(a)			
		Mean rank	<i>p</i> -value
NWP TCs versus NA TCs	NWP	424673	0.00
	NA	398196	
NWP TCs versus NEP TCs	NWP	470622	0.00
	NEP	356827	
NA TCs versus NEP TCs	NA	451889	0.00
	NEP	371556	
(b)			
		Mean rank	<i>p</i> -value
NWP TCs versus NA TCs	NWP	227414	0.00
	NA	180776	
NWP TCs versus NEP TCs	NWP	256067	0.00
	NEP	157066	
NA TCs versus NEP TCs	NA	224593	0.00
	NEP	184538	

Table S3. The results of the Mann-Whitney U test between the rainband-occurrence probabilities in different ocean basins. (a) The whole region of TCs and (b) the downshear region of TCs.

	25% quartile	Median	75% quartile	Tri-mean	RMS
Distances (km)	7.35	11.76	19.83	12.98	19.82
MWR-normalized distances (MWR)	0.19	0.33	0.59	0.37	0.71

Table S4. The statistics of the distances and MWR-normalized distances between the BT-interpolated and manually-recognized TC center positions.

Text S1.

There are 392 images covering TC eyes, which account for 84.48% of the total number (464) of all SAR TC images in the study. We calculated the distances between the BT-interpolated and manually-recognized TC center positions and normalized the distances with the RMWs. The histograms of the distances and RMW-normalized distances (i.e., distances divided by RMWs) are given in Figure S13, where magenta lines denote the positions of 25% quartile, 50% quartile (median), and 75% quartile. Table S4 presents the quartiles, tri-mean, and root mean squares (RMS) of the distances and RMW-normalized distances. The statistics in Table S4 are small to the considered radius range of two to ten times the MWR in the study.

Text S2.

We randomly chose 100 SAR TC images from the Northern Hemisphere's early morning and evening datasets as the testing dataset, respectively; thus, the number of randomly chosen SAR TC images is 200. We used the remaining 264 SAR TC images and their annotated data as the training and validation datasets for building the U-Net deep learning model. The loss function is the soft IoU (intersection over union) and the optimizer is the Adam algorithm.

Figure S14 shows three SAR TC images (Figures S14a, S14d, and S14g) in the testing dataset and the extracted rainbands by the manual annotation (Figures S14b, S14e, and S14h) and automatic extraction (Figures S14c, S14f, and S14i), respectively. Although the extracted rainbands are similar, the U-Net model wrongly recognizes the areas near the islands as rainbands in Figure S14c.

We used the IoU to evaluate the difference between the automatic extraction and manual annotation in the testing dataset. Its definition is:

$$\text{IoU} = \frac{|X \cap Y|}{|X \cup Y|} \quad (\text{S1})$$

where X and Y are the automatically extracted and manually annotated rainband areas here. The IoU of the testing dataset is 0.40, which means that although the automatic extraction results are consistent with the manual annotation overall, there remains much room for improving the U-Net model, requiring many samples.

Based on the automatically extracted rainbands in the testing dataset, the rainband-occurrence probability is calculated in Northern Hemisphere TCs, Northern Hemisphere early morning TCs, and Northern Hemisphere evening TCs, respectively, and given in Figures S15a to S15c. The radial distribution, downshear-region radial distribution, and azimuth distribution of Figures S15a to S15c are presented in Figures S16a to S16c, S17a to S17c, and S18a to S18c, respectively. The figures are consistent with those (Figures 3a to 3c, S7a to S7c, S8a to S8c, and S9a to S9c) of the manual annotation overall in variation. The rainband-occurrence probability based on the automatically extracted rainbands is slightly higher because the U-Net model wrongly recognized the non-rainband areas as the rainbands (Figure S14c gives an example where the U-Net model wrongly recognizes the areas near the islands as rainbands.) These misrecognized rainbands cause the overestimation of the rainband-occurrence probability.

Figures S15b and S15c of the automatic extraction and 3b and 3c of the manual annotation show similar peak positions (marked in magenta ellipse) of the rainband-occurrence probability. Both consistently indicate that the rainband-occurrence probability peak appears in the region farther from the TC center in the evening.

Based on the automatically extracted rainbands, we also checked the difference between the rainband-occurrence probability of Northern Hemisphere TCs in the early morning and evening. The early morning/evening rainband-occurrence probability difference was computed (Figure S19). Most grid nodes (53.67% of the grid nodes with difference values) have values greater than or equal to zero. The median of the rainband-occurrence probability for all grid nodes is 9.68% in the early morning and 9.09% in the evening. The Mann-Whitney U test indicates that the rainband-occurrence probability in

the early morning is statistically significantly higher than in the evening, with mean ranks of 425851 and 400761 and a p -value of 0.00. For the downshear region of TCs, the above values are 54.23%, 10.94%, 10.42%, 211398, 200902, and 0.00. Therefore, the results of the automatic extraction are also consistent with the conclusion in the study that rainbands in Northern Hemisphere TCs overall have a higher occurrence probability in the early morning.

Thus, the rainband-occurrence probability calculated from the automatically extracted rainbands also supports the findings in the study.